

Web Analytics and Reporting Setup Guide

GA4 configuration, custom event tracking, conversion measurement, and a monthly reporting template — built around Google Analytics 4, the only Google Analytics platform since Universal Analytics was sunset in 2023.

GA4's interface continues to evolve, and exact menu paths shift between updates. This guide focuses on what to configure and why, which stays valid across interface changes — check Google's current Analytics Help documentation for the exact click-path if a described setting isn't where you expect.

1. GA4 Setup Guide

GA4's structure: Account → Property → Data Stream. Get this hierarchy right from the start, since restructuring it later means losing historical continuity.

Step	What to Do
Create the property	One property per distinct business or app you want reported on separately; don't create one property per department if they're really the same business
Add a data stream	A web data stream for your site; add separate app streams if you also have mobile apps you want in the same property
Install tracking	Use Google Tag Manager rather than hard-coding the tracking snippet directly — it makes adding future tags and events far easier without a developer for every change
Configure data retention	Extend beyond the default where your reporting needs longer historical comparison windows; longer retention has no real downside beyond storage, which Google manages
Set up consent mode	Required for compliant tracking in regions with consent requirements (e.g. the EEA) — misconfigured consent settings can silently cut off audience and remarketing data for affected regions
Link related accounts	Connect Google Ads, Search Console, and BigQuery (if used) at setup time — linking later doesn't retroactively connect historical data

2. Custom Event Tracking

GA4 is built entirely around events, not pageviews as the primary unit — every interaction, including a pageview itself, is an event. Three tiers exist: automatically collected events (page views, scrolls), enhanced measurement events (file downloads, outbound clicks, enabled by default), and custom events you define yourself.

Planning Custom Events

- Start from the business action, not the technical action — ask “what does this user doing X actually mean for the business?” rather than tracking everything technically trackable
- Use consistent snake_case naming (e.g. generate_lead, start_trial) — inconsistent naming across a team makes later reporting genuinely painful
- Attach parameters that add real analytical value (e.g. which specific form was submitted), not just parameters for their own sake
- Test every new event in Realtime or DebugView before relying on it in reports — a silently broken event can go unnoticed for months and quietly invalidate a reporting period

Common Events Worth Tracking

Event	Typical Use Case
generate_lead	A contact form or quote request submission
sign_up	Account or newsletter registration
view_item / add_to_cart / purchase	E-commerce funnel stages (often available via enhanced e-commerce measurement)
file_download	Resource or document downloads (often covered by enhanced measurement automatically)
video_engagement	Meaningful video interaction, if video content matters to your goals

3. Conversion Measurement

Marking an event as a key conversion event tells GA4 (and connected tools like Google Ads) that this action represents real business value, not just user activity worth observing.

Keep the conversion list short and genuinely meaningful. A conversion list full of low-intent actions makes attribution noisy and can mislead budget decisions that rely on it — mark only the events that represent real outcomes, and track everything else as a regular event for analysis instead.

Setting Up Conversions

- Mark the specific events that represent a real outcome (a completed purchase, a qualified lead form, a trial signup) as key events
- For e-commerce, ensure the purchase event carries an accurate value and currency parameter — conversion value reporting depends entirely on this being correct
- Be aware of the attribution model applied to conversions — data-driven attribution (which distributes credit across the touchpoints that actually contributed) is now the standard default, replacing simpler last-click models
- Revisit the conversion list periodically — what counted as a meaningful action when the site launched may not still be the most meaningful action a year later

4. Monthly Reporting Template

Report Period: [Month, Year]

Headline Metrics

Metric	This Month	Last Month	Change
Users			
Sessions			
Key events (conversions)			
Conversion rate			
Average engagement time			

Traffic Sources

Channel	Sessions	Conversion Rate	Notable Change
Organic Search			
Paid Search			
Direct			
Referral			
Social			

Top Pages

[List top 5 pages by views or by conversions, whichever better serves this report's purpose]

Notable Changes and Annotations

Use GA4's Annotations feature to tag known events (a campaign launch, a site migration, a pricing change) directly on your reporting timeline — this makes future analysis far faster, since you won't need to cross-reference an external calendar to explain a spike or drop.

[Note any known campaigns, launches, or changes this period, and their likely effect on the numbers above]

Action Items for Next Period

[] [Action item based on this month's findings]

[] [Action item based on this month's findings]

5. Data Interpretation Guide

The most common analytics mistakes happen in interpretation, not in setup — the numbers can be perfectly accurate and still be read incorrectly.

Pitfall	What to Do Instead
Treating a single-session anomaly as a trend	Look for a pattern across multiple periods before concluding something meaningful changed
Confusing correlation with causation	A metric moving alongside a change doesn't confirm the change caused it — look for a plausible mechanism, not just timing
Over-interpreting data on very low-traffic pages or segments	Small sample sizes produce noisy percentages — a 50% conversion rate on 4 visitors isn't a meaningful signal
Comparing channel performance without accounting for attribution model	Different attribution models redistribute credit differently across channels — comparing across models without accounting for this produces misleading conclusions
Confusing users, sessions, and events	A single user can generate many sessions and many events — pick the metric that actually matches the question being asked
Chasing vanity metrics	Raw traffic or pageview growth alone doesn't confirm the business is doing better — tie reporting back to the key events that actually represent outcomes